PSCI-hat: Formal Spectral Bridge Hypothesis (Working Draft v0.1)

Status tier: HYPOTHESIZED throughout. Closes E1 ("bridge asserted not argued"). Not yet sent to Pellis.

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1. Scope — which operator, explicitly

Two distinct operators appear in the Pellis framework, with **opposite stability sign conventions**. This document is scoped to exactly one of them, stated up front to avoid the conflation flagged in stress-testing:

Operator	Source	Flow equation	Stability criterion	Adjoint type
$\hat{H}_P$	Theorem 10, second variation at equilibrium U^*	$\partial_t\, \delta P = -\hat{H}_P\, \delta P$	stable $\Longleftrightarrow$ $\lambda_n > 0\ \forall n$	self-adjoint
$\hat{L}_P$	Theorem 37, linearization along a trajectory	$\partial_t\, \delta\Phi = \hat{L}_P\, \delta\Phi$	stable $\Longleftrightarrow$ $\mathrm{Re}(\lambda_n) < 0\ \forall n$	**not** self-adjoint in general

This document is scoped to $\hat{L}_P$. Disease emergence (Theorem 37, eq. 2187) is defined on this operator, and it is the one any physiological/PSCI bridge needs to track.

This matters because the corpus's existing binding term $\mathbf{V(\psi) = \inf \mathrm{Re}(\lambda)}$ ("stability functional," retired alias "spectral floor") was calibrated to $\hat{H}_P$'s sign convention, where a *positive* infimum is the stable regime. Porting that definition directly onto $\hat{L}_P$ would invert the result. For $\hat{L}_P$, the correct stability-margin quantity is the **spectral abscissa**:

$$\alpha(t) := \sup_n \mathrm{Re}(\lambda_n(\hat{L}_P(t)))$$

stable $\Longleftrightarrow$ $\alpha(t) < 0$; criticality $\Longleftrightarrow$ $\alpha(t) = 0$; disease onset (eq. 2187) $\Longleftrightarrow$ $\alpha(t) > 0$.

$\alpha(t)$ is a standard object in linear stability/control theory (it is the rightmost point of the spectrum) and is the direct, sign-correct analog of $V(\psi)$ for this operator.

This is distinct from Pellis's own $\mathbf{\Delta(t)}$ (eq. 2188, $\min_{i\neq j}|\mathrm{Re}(\lambda_i)-\mathrm{Re}(\lambda_j)|$, mode-spacing). $\alpha(t)$ answers "how close to the instability threshold." $\Delta(t)$ answers "how close two modes are to each other." Conflating these was the error flagged in stress-testing; they are kept separate here.

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2. Definition: PSCI-hat

`PSCI-hat` occupies the existing, named-but-unfilled slot in the controlled vocabulary ("spectral collapse index, surrogate"). It is **not** a redefinition of `D(t)`; `D(t)` remains purely coherence-defined (`D(t) = [C0−C(t)]/C0`), per the binding spec.

```
PSCI-hat(t) := σ( α(t) / α_ref )
```

where:

- `σ` is any bounded, strictly monotonic function with `σ(0) = 0.5` (e.g. logistic or `(1+tanh(x))/2`) — the specific shape is a **disclosed modeling choice**, not a derived necessity.
- `α_ref` is a reference rate (same units as `α`, i.e. inverse time), required for dimensional consistency and pending reconciliation with the Prong 2 (REAL) reference-scale work.
- `PSCI-hat(t) = 0.5` exactly at `α(t) = 0` — i.e., at Pellis's own disease-onset threshold (eq. 2187), by construction, for any valid `σ` and `α_ref`.

Assumption stated explicitly: this requires `L̂_P(t)` to have discrete spectrum (so `sup_n Re(λ_n)` is attained, not merely an unattained supremum). This is not automatic on a non-compact, time-evolving manifold and has not been verified against Pellis's §2 manifold definition. Flagged as an open item, not assumed away.

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3. The bridge hypothesis (closes E1)

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H_bridge:  D(t) and PSCI-hat(t) share the same critical functional form
           as their respective systems approach threshold, within
           equivalence margin ε_eq.
```

This is the explicit, falsifiable statement that E1 found missing. It is tested by the gates that already exist in the corpus:

- **Q1 (decay-form gate):** does `D(t)`'s approach to breakdown match `PSCI-hat(t)`'s approach to `α(t)=0` in functional shape (not just qualitatively, but in decay/rise form)?
- **Q2 (exponent correspondence gate):** do the two critical exponents match within `ε_eq`?

Q2 is where the exceptional-point finding sharpens the prediction rather than just renaming it.

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4. The exponent prediction (the actual contribution to Pellis's math)

`L̂_P` is non-self-adjoint. Degeneracies of non-self-adjoint operators are **exceptional points**, not the diabolical points / avoided crossings that govern self-adjoint degeneracies — a distinction the paper does not currently draw between Theorem 10 and Theorem 37.

At an exceptional point, eigenvalues *and* eigenvectors coalesce, and the local behavior of the eigenvalue near the degeneracy is a **square-root branch point**: as a control parameter `ε` is swept through its critical value `ε_c`,

```
α(ε) ≈ α_c ± k·√(ε - ε_c)
```

i.e. a critical exponent of **1/2** — sharper than a generic (linear, exponent-1) crossing.

Falsifiable prediction for his own simulation: if the degeneracy underlying Theorem 37's bifurcation is genuinely an exceptional point of `L̂_P`, his own **S2 criterion ("exponent recovery within tolerance")** should recover an exponent of `1/2`. If the simulated data instead recovers exponent `1`, the degeneracy is a regular (non-exceptional) crossing, and the EP hypothesis is falsified — which is itself a useful result, not a failure of this document.

This also gives a precise, non-hand-wavy account of why eq. 2186's modal expansion (`δρ = Σ aₖ(t)Ψₖ(x)`) breaks down near `Δ(t)→0`: at an exceptional point, `{Ψₖ}` is not merely closely spaced, it is **defective** — geometric multiplicity falls below algebraic multiplicity, and the basis genuinely fails to span the space. This replaces the earlier, vaguer "degenerate eigenvalues are noisy" framing with a specific, checkable mechanism.

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5. What this resolves vs. what remains open

Resolved by this document (no dependency on Pellis):

- E1 — bridge is now a stated hypothesis with a testable functional form, not an assertion
- Operator ambiguity — `L̂_P` named explicitly, sign convention corrected, distinguished from `Ĥ_P`
- `Δ(t)` vs. `α(t)` conflation — separated into two named, non-overlapping quantities
- Vague degeneracy language — replaced with the specific EP mechanism and a numbered exponent prediction

Still open, dependent on Pellis's simulation (Prong 3 territory):

- Whether `L̂_P` actually has discrete spectrum on the biological manifold (§2 assumption, unverified)
- The actual numerical value of the recovered exponent (Q2) — this is the test, not something to assume in advance
- Type I/II calibration of any PoR/RCR gate built downstream of `PSCI-hat` — still requires real eigenvalue data to compute

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6. Disclosure note for any external version

If/when this is shared outside this working draft: the `σ` function and `α_ref` choice in §2 must be labeled as a modeling convention, not a derived result, exactly as flagged in §2. The exponent prediction in §4 must be labeled as a hypothesis his own S2 test can confirm or falsify — not as an established correspondence.